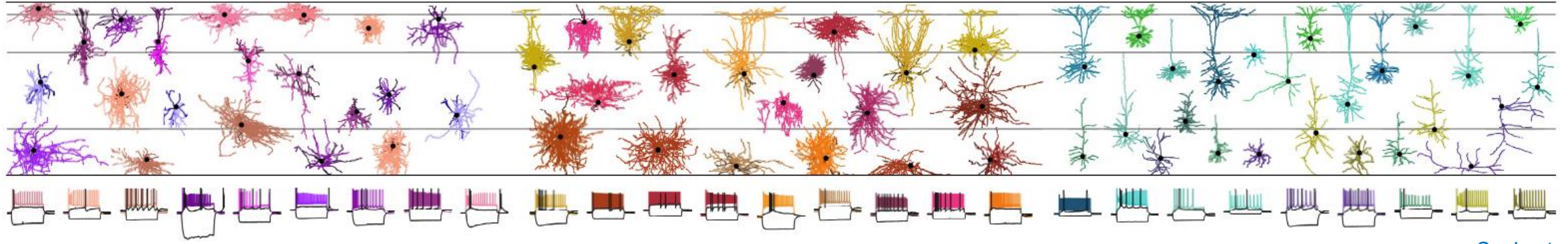


Epilepsy Case Study: Mapping Human Neuron Diversity in Search for New Therapies



[Scala et al. \(2021\)](#)

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Pch.vector / freepik

CASE STUDY

Mapping Human Neuronal Diversity in the Search for New Therapeutics: Using Real Human Neuron Data Sets to Build Student Quantitative Skills

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Case Study: Content Learning Objectives

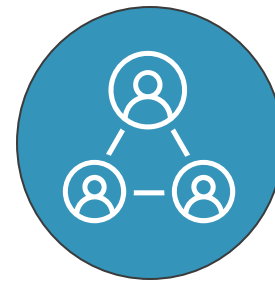
You should be able to...

1. Describe epilepsy and how seizures occur in the brain
 - a. Compare different types of seizures
 - b. Analyze new evidence that reveals how acute focal seizures spread in the brain
 - c. Explain SUDEP and when to call 911 if someone is having a seizure
2. Discuss how neuronal diversity may impact the therapeutic potential of new drugs
3. Define neuronal excitability and identify factors that influence it
4. Discuss how epilepsy linked mutations and anti-epileptic drugs can alter neuronal excitability
5. Access human neuron electrophysiological, morphological and transcriptomic data from epileptic patients
6. Analyze electrophysiological data from human epileptic patients
7. Discuss the ethical, legal and social implications of this human neuron data set
8. Discuss the advantages and limitations of studying neurons from human patients

Technical Skills Based on Learning Objectives

You should be able to...

1. Apply big data from high-throughput experiments to detect patterns and quantify data
2. Identify and test a new hypothesis
3. Apply appropriate statistics to quantitative results
4. Describe practical considerations of data collection in humans
5. Access data from high-throughput experiments on living human brain tissue in the Allen Institute database
6. Interpret primary science data in the context of pre-translational research



Cell Types Dataset: <https://celltypes.brain-map.org/>

Cell Feature Search

The brain cell database is a survey of biological features measured from single cells, in human and mouse. Explore cell properties and access detailed experimental data using the feature browser below.

DOWNLOAD CELL FEATURE DATA

Species

☐ Human ☐ Mouse

Location

Select Areas

Select Layers

Hemisphere

☐ Left ☐ Right

Transgenic Targeting

Select Mouse Lines

Reporter status

☐ Positive ☐ Negative

Donor Profile

Age

1890

Sex

☐ Female ☐ Male

Years of seizure

060

Select Disease

Select Donor

Ethnicity/Race

Morphology annotation

Reconstruction type

☐ Full ☐ Dendrite Only ☐ None

Dendrite type

☐ Spiny ☐ Aspiny ☐ Sparsely spiny

Apical dendrite status

☐ Intact ☐ Truncated ☐ N/A

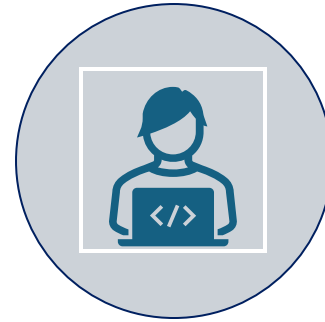
Models

☐ Has GLIF Model ☐ Has All-active Biophysical model ☐ Has Perisomatic Biophysical model

Outline of Topics Today



Introduce learning objectives & case study



Code along demo



Demonstrate ways to implement in classes



Share open access resources/materials

Introduction to the Case

You have just started work in the lab of Dr. Naomi Marsh in the neurology department at the university hospital. Dr. Marsh is conducting research on a new drug that is thought to affect neuronal excitability and she wants to test its effect on the properties of temporal lobe neurons. Temporal lobe epilepsy is the most common form of focal epilepsy, so she is especially interested in testing this compound's effects on cells in that region of the brain.

Dr. Marsh hopes the drug will affect the properties of temporal lobe neurons to reduce the likelihood of a seizure. However, first she needs to gather some baseline data on temporal lobe neurons' excitability, and how these cells may be distinct from cells in other areas of the brain.

You have access to living neurons' data from tissue samples collected from neurosurgery patients for the Allen Cell Types Database, a resource by the Allen Institute for Brain Science. The team examines tissue removed as part of routine neurosurgery for epilepsy and glioblastoma. The tissue included in this dataset is in the path the surgeons have to take to access the pathological site. The seizure focus or tumor from the patient are examined by a pathologist and not included in this dataset. The team has measured these cells' properties and shared them in an open database that you can use to do this experiment.

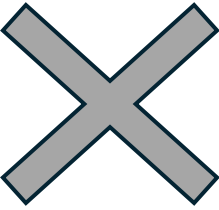
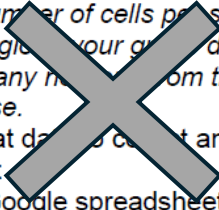
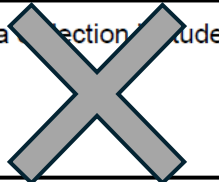
Original Format

	Day 1	Day 2
Pre-Work	1. Read “The Seizures of Epilepsy” from <i>Neuroscience Exploring the Brain</i> 2. Watch 2-min Neuroscience: Epilepsy Video 3. Read “Introduction to the case” and “Experimental design of the Allen Cell Types Database and your experiment” in the case study document. 4. Answer the “Pre-work Questions” in the case study.	1. Ensure data collection is complete. 2. OPTIONAL: Begin data analysis with group
In-Class: Background Material and Introduction	Instructor introduces related topics: 1. Seizures vs. epilepsy 2. Types of seizures 3. How seizures spread in the brain 4. Neuronal diversity 5. Allen institute and local neurosurgeon partnerships	Instructor introduces related topics: 1. Neuronal excitability (firing rates, resting potentials, etc.) 2. Seizures and neuronal excitability 3. Explain SUDEP and what to do in a seizure emergency.
In-Class: Data Collection and Analysis	Students collect data as a class using the Google sheet 1. Assign ~10 cells per student for data collection. <i>Note the number of cells per student depends on the brain regions your group decides to explore and how many neurons from those regions are in the database.</i> 2. Explain what data to collect and the organization of the sheet 3. A sample Google spreadsheet can be found in supplemental materials (Supplementary Material 2).	In small groups (4 students) analyze class data from the Google Sheet 1. Calculating means and standard deviation for the firing rate and resting potential for each brain region and cortical layer within the region 2. Compare two regions (MTG and FroL) with a two-sample Wilcoxon/Mann-Whitney U-test. 3. Discuss why a U-test and not T-test 4. Interpret the results or the statistical comparison. 5. Ask a new research question and design a way to test the new hypothesis, show results with a table or graph, and interpret the statistical comparison.
Post-Work	Complete data collection if students did not finish during class.	Complete unfinished data analysis and the post-data analysis questions in the case study (Supplementary Material 1).

[Milligan et al. 2023 \(JUNE\)](#)

Table 1. Case implementation plan broken down into pre-class, in-class, and post-class work for two 50-minute class session.

Example Modifications

	Day 1	Day 2
Pre-Work	1. Read "The Seizures of Epilepsy" from <i>Neuroscience Exploring the Brain</i> 2. Watch 2-min Neuroscience: Epilepsy Video 3. Read "Introduction to the case" and "Experimental design of the Allen Cell Types Database and your experiment" in the case study document. 4. Answer the "Pre-work Questions" in the case study.	1. Ensure data collection is complete. 2. OPTIONAL: Begin data analysis with group 
In-Class: Background Material and Introduction	Instructor introduces related topics: 1. Seizures vs. epilepsy 2. Types of seizures 3. How seizures spread in the brain 4. Neuronal diversity 5. Allen institute and local neurosurgeon partnerships	Instructor introduces related topics: 1. Neuronal excitability (firing rates, resting potentials, etc.) 2. Seizures and neuronal excitability 3. Explain SUDEP and what to do in a seizure emergency.
In-Class: Data Collection and Analysis	Students collect data as a class using the Google sheet 1. Assign ~10 cells per student for data collection. <i>Note the number of cells per student depends on the brain region your group decides to explore and how many regions from those regions are in the database.</i> 2. Explain what data is collected and the organization of the sheet 3. A sample Google spreadsheet can be found in supplemental materials (Supplementary Material 2). 	In small groups (4 students) analyze class data from the Google Sheet 1. Calculating means and standard deviation for the firing rate and resting potential for each brain region and cortical layer within the region 2. Compare two regions (MTG and FroL) with a two-sample Wilcoxon/Mann-Whitney U-test. 3. Discuss why a U-test and not T-test 4. Interpret the results of the statistical comparison. 5. Ask a new research question and design a way to test the new hypothesis, show results with a table or graph, and interpret the statistical comparison.
Post-Work	Complete data collection. Students did not finish during class. 	Complete unfinished data analysis and the post-data analysis questions in the case study (Supplementary Material 1).

[Milligan et al. 2023 \(JUNE\)](#)

Table 1. Case implementation plan broken down into pre-class, in-class, and post-class work for two 50-minute class session.

Modified Versions for 2 Courses

Similar Activities Across My Courses

Pre-Work	1. Read “The Seizures of Epilepsy” from Neuroscience Exploring the Brain 2. Watch 2-min Neuroscience: Epilepsy Video 3. Read “Introduction to the case” and “Experimental design of the Allen Cell Types Database and your experiment” in the case study document 4. Answer the “Pre-work Questions” in the case study
In-Class: Background Material & Introduction	Instructor introduces topics: 1. Seizures vs. epilepsy 2. Types of seizures 3. How seizures spread in the brain 4. Neuronal diversity 5. Allen institute and local neurosurgeon partnerships 6. Neuronal excitability (firing rates, resting potentials, etc.) 7. Seizures and neuronal excitability 8. Statistics prep and discussion *Combined into 1 class session (~1 hr)

	Neurotechniques Laboratory Course (1x per week / 4 hrs)	Data Science for Neuroscience (Intro to Coding) (2x per week / 75 mins)
*Data Analysis (Collection/organization skipped)	Instructor leads + small group work in code notebook: 1. Calculating means and standard deviation for the firing rate and resting potential for each brain region and cortical layer within the region 2. Compare two regions (MTG and FroL) with appropriate statistics 3. Discuss using a U-test vs T-test 4. Interpret the results of the statistical comparison 5. Ask a new research question and test the new hypothesis, show results with plots, and interpret the statistical comparison	• Code notebook completed as homework • Minimal scaffolding + solutions omitted
Post-Work	Complete data analysis notebook and *submit short lab report write-up (emphasis on methods & results)	Complete data analysis notebook and post-data analysis questions in case study document

***indicates differences from published example**

Code Notebook Shared From Lab Course

Pre-Work	1. Read “The Seizures of Epilepsy” from Neuroscience Exploring the Brain 2. Watch 2-min Neuroscience: Epilepsy Video 3. Read “Introduction to the case” and “Experimental design of the Allen Cell Types Database and your experiment” in the case study document 4. Answer the “Pre-work Questions” in the case study
In-Class: Background Material & Introduction	Instructor introduces topics: 1. Seizures vs. epilepsy 2. Types of seizures 3. How seizures spread in the brain 4. Neuronal diversity 5. Allen institute and local neurosurgeon partnerships 6. Neuronal excitability (firing rates, resting potentials, etc.) 7. Seizures and neuronal excitability 8. Statistics prep and discussion *Combined into 1 class session (~1 hr)

	Neurotechniques Laboratory Course (1x per week / 4 hrs)
*Data Analysis (Collection/organization skipped)	Instructor leads + small group work in code notebook: 1. Calculating means and standard deviation for the firing rate and resting potential for each brain region and cortical layer within the region 2. Compare two regions (MTG and FroL) with a two-sample Wilcoxon/Mann-Whitney U-test 3. Discuss why a U-test and not T-test 4. Interpret the results of the statistical comparison 5. Ask a new research question and test the new hypothesis, show results with plots, and interpret the statistical comparison
Post-Work	Complete data analysis notebook and *submit short lab report write-up (emphasis on methods & results)

Considerations

- Students have minimal background in coding
 - Not explicitly a coding class
- Instructor led code-along session
 - Scaffolding for independence
 - Most solutions included
- Final notebook section - group/partner work
- Submitted for grading:
 - Code notebook
 - Short lab report
 - Emphasizes results, interpretation, & figures

***indicates differences from published example**

EpilepsyCaseStudy_CellTypes_Notebook.ipynb

File Edit View Insert Runtime Tools Help

Commands Code Text Run all Copy to Drive

Table of contents

Epilepsy Case Study Notebook

Before you start - Save this notebook!

Learning Objectives

Survey

1.0 Setup, import, and load files needed

1.1 Setup

Task

Task

Task

Alert

1.2 Import packages needed

Task

1.3 Restarting your notebook

1.4 Load in the data file

Task

2.0 Viewing the Data

Task

DataFrames

DataFrame Index

DataFrames versus Series

2.1 Preview the data and columns

Task

Task

2.1.1 Formatting the dataframe - many options!

2.2 Data Column Details:

2.3 Viewing all columns in the dataframe

Open in Colab

Epilepsy Case Study Notebook

Investigating Questions of Interest Using the Allen Institute Cell Types Dataset

It's time to access the Allen Cell Types Database to do some analysis for Dr. Marsh's research. Refer to the Case Study Document posted on Webcampus for details on how this dataset was collected and how to interpret it.

This notebook will help us investigate data collected from the [Cell Types](#) dataset from the Allen Brain Institute.

IMPORTANT: This notebook is READ-ONLY. To edit and run this notebook, follow the steps below.

Before you start - Save this notebook!

When you open a new Colab notebook from the WebCampus (like you hopefully did for this one), you cannot save changes. So it's best to store the Colab notebook in your personal drive "File > Save a copy in drive..." **before** you do anything else.

The file will open in a new tab in your web browser, and it is automatically named something like: "Copy of **EpilepsyCaseStudy_CellTypes_Notebook.ipynb**". You can rename this to just the title of the assignment "EpilepsyCaseStudy_CellTypes_Notebook.ipynb". Make sure you do keep an informative name (like the name of the assignment) so that you know which files to submit back to WebCampus for grading! More instructions on this are at the end of the notebook.

Where does the notebook get saved in Google Drive?

By default, the notebook will be copied to a folder called "Colab Notebooks" at the root (home directory) of your Google Drive. If you use this for other courses or personal code notebooks, I recommend creating a folder for this course and then moving the assignments AFTER you have completed them.

Learning Objectives

At the end of this assignment, you'll be able to:

- Analyze electrophysiological data from human epileptic patients
 - Experimental data was collected from living human brain tissue from the Allen Institute
 - Apply big data from high-throughput experiments to detect patterns and quantify data
- Apply appropriate statistics to quantitative results
- Describe practical considerations of data collection in humans
- Interpret basic science data in the context of pre-translational research

https://github.com/AllenInstitute/EducationAndEngagement/blob/main/NeuroCodingWorkshop/EpilepsyCaseStudy_CellTypes_Notebook.ipynb

Epilepsy Case Study Cell Types Notebook

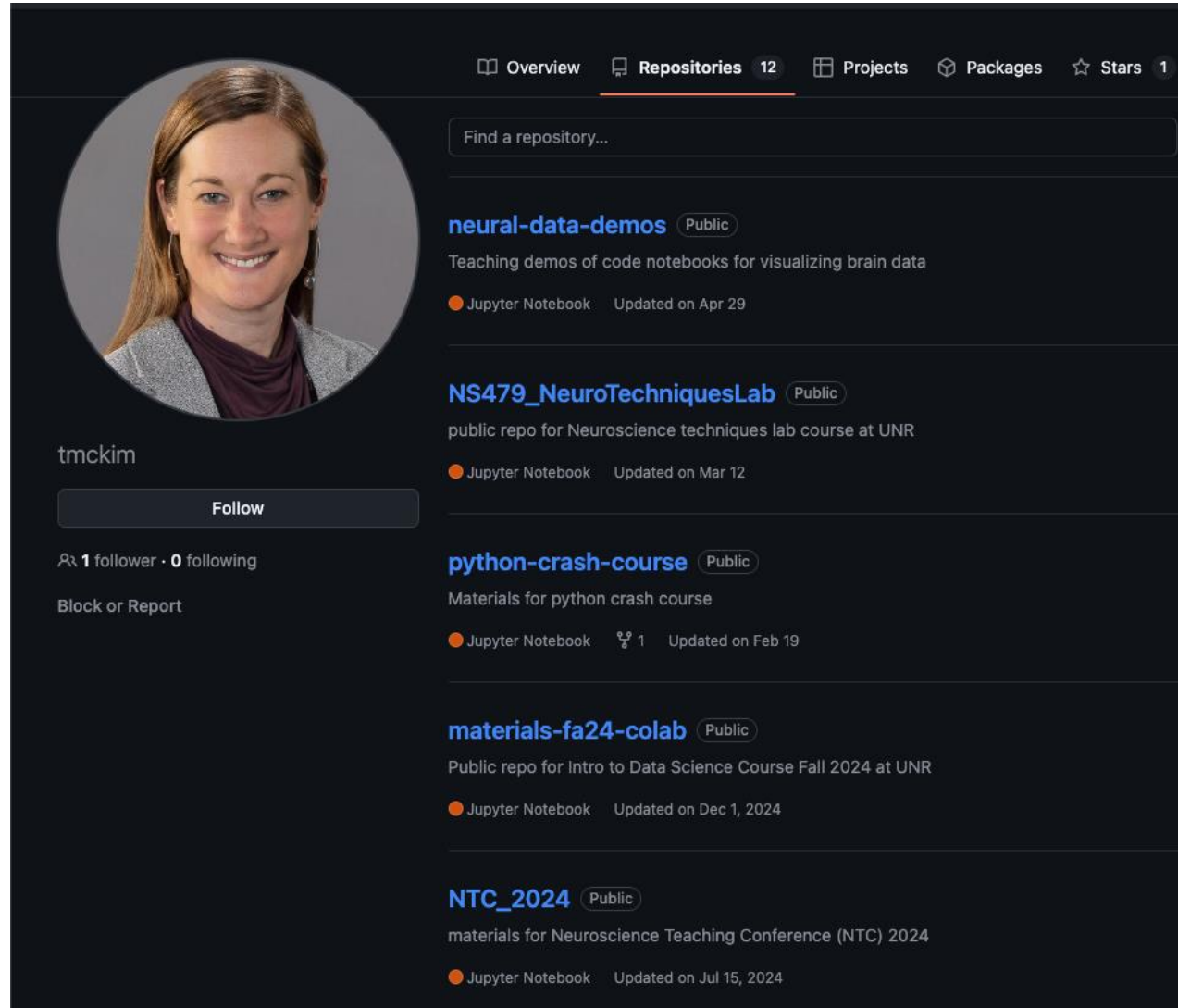
Opening the notebook:

1. Click/copy the link to github
2. Click on  Open in Colab

Contact me if you'd like materials/resources:
tmckim@unr.edu

- Open-ended notebook for coding class includes questions + points breakdown for grading
- Solutions for different hypotheses they can test (+ visualizations)
- Example lab writeups, etc.
- Pre/post surveys on interest, outcomes, etc.

Other resources @ github.com/tmckim



The screenshot displays the GitHub profile of user **tmckim**. The profile includes a circular profile picture of a woman with long brown hair, the username **tmckim**, and a **Follow** button. Below the profile information, it shows **1 follower** and **0 following**, along with a **Block or Report** link. The **Repositories** tab is selected, showing a list of five public repositories. Each repository entry includes the repository name, its description, the type of content (Jupyter Notebook), and the last update date.

Repository Name	Visibility	Description	Content Type	Last Updated
neural-data-demos	Public	Teaching demos of code notebooks for visualizing brain data	Jupyter Notebook	Updated on Apr 29
NS479_NeuroTechniquesLab	Public	public repo for Neuroscience techniques lab course at UNR	Jupyter Notebook	Updated on Mar 12
python-crash-course	Public	Materials for python crash course	Jupyter Notebook	Updated on Feb 19
materials-fa24-colab	Public	Public repo for Intro to Data Science Course Fall 2024 at UNR	Jupyter Notebook	Updated on Dec 1, 2024
NTC_2024	Public	materials for Neuroscience Teaching Conference (NTC) 2024	Jupyter Notebook	Updated on Jul 15, 2024